



STANDARD OPERATING PROCEDURE (SOP)

End-to-End E-Commerce Analytics Platform

Using PostgreSQL and Power BI



**Data
Management**



**Data
Processing**



**Business
Intelligence**



**Insights &
Decision Making**

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Dataset:

Olist E-Commerce Dataset
(CSV Files)

STANDARD OPERATING PROCEDURE (SOP)

1. Purpose

The purpose of this Standard Operating Procedure (SOP) is to define the complete process followed for developing the End-to-End E-Commerce Analytics Platform. This document provides guidelines for data acquisition, database setup, data validation, business intelligence view creation, Power BI integration, dashboard development, and performance optimization. Following this procedure ensures data accuracy, reporting consistency, and efficient business intelligence generation.

2. Dataset Acquisition

The project uses the **Olist E-Commerce Dataset**, which was downloaded from Kaggle in ZIP format. After downloading, the ZIP file was extracted to obtain all required CSV files containing customer, order, product, seller, payment, review, and category information.

The extracted files served as the primary data source for the analytics platform.

3. Database Setup

A PostgreSQL database named **olist_db** was created to store and manage the e-commerce data. Separate raw tables were created for each CSV file, including customers, orders, order items, products, sellers, payments, reviews, category translation, and other supporting datasets.

After table creation, all CSV files were imported into PostgreSQL using the COPY command and pgAdmin import utilities. This ensured that the raw data was available for further processing and analysis.

4. Data Validation and Quality Assurance

After loading the data into PostgreSQL, several validation checks were performed to ensure data quality and integrity. These checks included verifying row counts, checking for null values, validating primary key uniqueness, reviewing sample records, and ensuring proper relationships between tables.

Data validation was essential to confirm that the imported data matched the source dataset and could be safely used for reporting and analysis.

5. Database Optimization

To improve query performance and reporting efficiency, indexes were created on frequently used columns such as **order_id**, **customer_id**, **product_id**, **seller_id**, and **purchase_timestamp**. These indexes reduced query execution time and improved dashboard responsiveness when Power BI accessed the database through DirectQuery mode.

6. Business Intelligence View Creation

Business Intelligence (BI) views were developed to simplify reporting and provide business-ready datasets. These views combined and transformed data from multiple raw tables into optimized analytical structures.

The primary reporting view created was **bi_fact_sales**, which serves as the central fact table for business analysis. Additional dimension views were also created to support product, customer, seller, and time-based analysis.

The BI views were tested extensively using SQL queries to verify record counts, eliminate duplicate records, and validate business calculations before connecting them to Power BI.

7. Power BI Integration

Power BI Desktop was connected directly to PostgreSQL using the DirectQuery connection mode. Instead of importing raw tables, only optimized BI views were loaded into Power BI to improve performance and simplify data modeling.

A separate **DimDate** table was created and linked to the purchase date field to support time-intelligence calculations such as Year-to-Date (YTD), Month-to-Date (MTD), and Year-over-Year (YoY) analysis.

Relationships between fact and dimension tables were configured and validated to ensure accurate reporting and filtering behavior.

8. KPI and Dashboard Development

DAX measures were developed to calculate key business metrics and performance indicators. These measures included Revenue, Orders, Customers, Average Order Value (AOV), Year-over-Year Growth, On-Time Delivery Percentage, and Average Customer Rating.

Using these KPIs, multiple report pages were developed to provide comprehensive business insights, including:

- Executive Overview Dashboard
- Sales Analysis Dashboard
- Product Analysis Dashboard
- Customer Analysis Dashboard
- Logistics Analysis Dashboard
- Reviews Analysis Dashboard
- Payments Analysis Dashboard
- Seller Analysis Dashboard

9. Performance Optimization

To ensure smooth dashboard performance, several optimization techniques were implemented. High-cardinality fields were minimized in visuals, complex calculations were optimized, unnecessary cross-filtering was reduced, and Power BI Performance Analyzer was used to identify bottlenecks.

Database indexes and optimized BI views further improved query performance, ensuring efficient data retrieval through DirectQuery mode.

10. Conclusion

The End-to-End E-Commerce Analytics Platform was successfully developed by following a structured workflow that included dataset acquisition, database setup, data validation, performance optimization, business intelligence view creation, Power BI integration, KPI development, and dashboard reporting. The resulting solution provides a centralized analytics platform that enables business users to analyze sales, customers, products, logistics, payments, reviews, and seller performance through interactive dashboards and actionable insights.